HULL LONGITUDINAL STRENGTH & MIDSHIP MODULUS BOOKLET

STATUTORY RULES 209 & 223 / SCHEDULE TABLES 66 & 67 / ISO 1976 INLAND HULL STRENGTH STANDARD

Vessel Name:	M. V. YOUNUS MOLLA-1	Operating Route:	Inland & bay crossing
Vessel Type:	Steel Body Cargo Vessel	Plying Time:	Day & Night
Hull Construction:	Single Hull	Statutory Factors kt / kz:	1.40 / 1.40
Length Over All (LOA):	76.21 m	Breadth Moulded (B):	15.24 m
Length LWL:	74.29 m	Depth Moulded (D):	4.87 m
Length BP (LBP):	69.85 m	Scantling Draft (d):	4.000 m
Block Coefficient (Cb):	0.750	Service Speed:	14 Knots
Deadweight (DWT):	2000 T	Displacement:	3006 T
Transverse Frame Spacing:	550 mm	Statutory Status:	COMPLIANT

EXECUTIVE STATUTORY & STRUCTURAL COMPLIANCE SUMMARY

Vessel 'M. V. YOUNUS MOLLA-1' satisfies all longitudinal strength criteria per Bangladesh Inland Shipping Rules 2001 (Rules 209 & 223), Schedule Tables 66 & 67, and ISO 1976 Inland Hull Strength Standard.

- Governing Design Bending Moment: 29,160.53 kN·m (SAGGING) | Design Shear Force: 1,043.68 kN
- Actual Provided Section Modulus: 563,428.07 cm³ exceeds Statutory Requirement 264,150.77 cm³ by +113.3% Margin.
- Maximum Hull Girder Continuous Bending Stress: 51.76 N/mm² (Allowable limit: 130.0 N/mm² per Schedule Table 66).

PREPARED & SUBMITTED BY:



Naval Architect & Marine Structural Consultant
Registration / Membership Seal

REVIEWED & STATUTORILY CHECKED BY:



Principal Engineer & Ship Surveyor
Government Approved Verification Seal

103 Calculation as per Statutory Rule 209 & ISO 1976 Standard

Statutory Particulars & Navigation Factor (Rule 209 & Table 66)

Rule Length, L (Length Between Perpendiculars, Lbp)	=	**69.85 m**
Summer Load Waterline Length, LWL	=	**74.29 m**
Moulded Breadth, B | Moulded Depth, D	=	**15.24 m | 4.87 m**
Scantling Draft, d | Block Coefficient, Cb	=	**4.000 m | 0.750**
Route Navigation Factor, kt (Inland / Coastal Plying Factor)	=	**1.40**

Statutory Wave Coefficient (Rule 209 & ISO 1976 Standard)

Wave Coefficient, Cw = $C_w = 0.08 \cdot k_t \cdot \sqrt{L}$

= **0.9361**

Still Water Longitudinal Bending Moment (Rule 209 Standard)

Still Water Bending Moment, Msw = $M_{sw} = 0.06 \cdot C_b \cdot L^2 \cdot B \cdot (d + 0.02 \cdot L)$

= **18,058.54 kN·m**

Dynamic Wave Bending Moments (Rule 209 & ISO 1976 Standard)

Wave Bending Moment, Hogging Condition, Mw,hog = $M_{w,hog} = 0.19 \cdot C_w \cdot L^2 \cdot B \cdot C_b$

= **9,918.70 kN·m**

Wave Bending Moment, Sagging Condition, Mw,sag = $M_{w,sag} = 0.11 \cdot C_w \cdot L^2 \cdot B \cdot (C_b + 0.70)$

= **11,101.99 kN·m**

Total Design Bending Moments & Hull Girder Shear Force

Total Longitudinal Bending Moment, Hogging, Mtotal,hog	=	**27,977.25 kN·m**
Total Longitudinal Bending Moment, Sagging, Mtotal,sag	=	**29,160.53 kN·m**
Governing Design Bending Moment, Mdesign	=	**29,160.53 kN·m (SAGGING)**

Hull Girder Maximum Design Shear Force, Qdesign = $Q_{design} = \frac{2.5 \cdot M_{design}}{L}$

= **1,043.68 kN**

Required Midship Section Modulus & Moment of Inertia (Rule 209)

Required Section Modulus by Stress Limit, Z1 (σallow = 130 N/mm²) = $Z_1 = \frac{M_{design} \cdot 10^3}{\sigma_{allowable}}$

= **224,311.8 cm³ (0.224312 m³)**

Required Section Modulus by Empirical Formula, Z2 = $Z_2 = 1.75 \cdot k_t \cdot L^2 \cdot B \cdot (C_b + 0.70)$

= **264,150.8 cm³ (0.264151 m³)**

Governing Statutory Required Section Modulus, Zreq = **264,150.8 cm³ (0.264151 m³)**

Minimum Hull Girder Moment of Inertia, Ireq = $I_{req} = 3.0 \cdot Z_{req} \cdot L \cdot 10^{-2}$

= **553,528 cm⁴ (0.005535 m⁴)**

Permissible Continuous Bending Moment, Mperm = **73,245.65 kN·m**

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

104 Calculation as per Design [Assumed Neutral Axis = 2678.5 mm ABL]

Items	Scantlings (mm)	Length (mm)	Breadth (mm)	Area (mm²)	Weight (Tonne)	Lever (mm)	Moment (mm³)	Area X Lever² (mm■)	I about own N.A. (mm■)
Items Above Assumed N.A.									
Sheer Strake (Port & Stbd)	PL-1000x12.0	1000	12.0	24000	13.16	1691.5	40,596,000	68,668,134,000	2,000,000,000
Strength Deck Plating (Abreast of Hatches)	6858x9.0	6858	9.0	61722	33.84	2187.0	134,986,014	295,214,412,618	416,624
Continuous Hatch Side Coamings	PL-600x13.0	600	13.0	15600	8.55	2491.5	38,867,400	96,838,127,100	468,000,000
Continuous Deck Girders Web (P & S)	PL-300x11.0	300	11.0	6600	3.62	2032.5	13,414,500	27,264,971,250	49,500,000
Continuous Deck Girders Flange (P & S)	175x11.0	175	11.0	3850	2.11	1877.0	7,226,450	13,564,046,650	38,821
Sub Total Above N.A.				**111,772**			**235,090,364**	**501,549,691,618**	**2,517,955,444**
Items Below Assumed N.A.									
Keel Plate	1100x12.0	1100	12.0	13200	7.24	-2672.5	-35,277,000	94,277,782,500	158,400
Bottom Shell Plating (Port & Stbd)	12540x12.0	12540	12.0	150480	82.51	-2672.5	-402,157,800	1,074,766,720,500	1,805,760
Bilge Plating (Port & Stbd Strake)	1257x14.0	1257	14.0	35185	19.29	-2387.8	-84,014,265	200,609,263,018	574,685
Side Shell Plating (Port & Stbd)	PL-3070x12.0	3070	12.0	73680	40.40	-343.5	-25,309,080	8,693,668,980	57,868,886,000
Center Keelson Web Plate	PL-400x12.0	400	12.0	4800	2.63	-2466.5	-11,839,200	29,201,386,800	64,000,000
Center Keelson Face Plate	250x12.0	250	12.0	3000	1.64	-2260.5	-6,781,500	15,329,580,750	36,000
Side Keelsons Web Plates (P & S)	PL-400x11.0	400	11.0	8800	4.83	-2466.5	-21,705,200	53,535,875,800	117,333,333
Side Keelsons Face Plates (P & S)	100x11.0	100	11.0	2200	1.21	-2261.0	-4,974,200	11,246,666,200	22,183
Sub Total Below N.A.				**291,345**			**-592,058,245**	**1,487,660,944,548**	**58,052,816,362**
Total Area / Inertia				**403,117**				**1,989,210,636,166**	**60,570,771,806**



Distance of neutral axis from assumed axis, y = $y = \frac{\sum M_{net}}{\sum A}$

= **885.5197 mm below**

Moment of Inertia about Assumed Axis, 1/2 Iassumed = $I_{assumed} = \sum (A \cdot y^2 + I_0)$

= 2.050e+12 mm⁴

Moment of Inertia about Actual N.A., 1/2 INA = $I_{NA} = I_{assumed} - A \cdot y^2$

= 1.734e+12 mm⁴

Total Hull Girder Moment of Inertia, INA (IY) = **3.467e+12 mm⁴**

= **1.7336790 m⁴ [> Ireq = 0.005535 m⁴]**

Height of Neutral Axis above Baseline, N.A. (ABL) = **1793.0 mm (1.7930 m)**

Rule 209 & ISO 1976 Statutory Modulus Evaluation

Section Modulus at Bottom Plating, ZAB = $Z_{AB} = \frac{I_{NA}}{y_{bottom}}$

= **0.966926 m³ (966,926.0 cm³) [> Zreq = 264,150.8 cm³]**

Distance from Neutral Axis to Deck, VD = **3.0770 m (3077.0 mm)**

Section Modulus at Continuous Deck, ZAD = $Z_{AD} = \frac{I_{NA}}{y_{deck}}$

= **0.563428 m³ (563,428.0 cm³) [> Zreq = 264,150.8 cm³]**

Continuous Longitudinal Steel Weight (Midship Structural Core)	**221.03**	**Tonne**
Transverse Framing, Bulkheads & Outfitting Allowance (56% Standard)	123.78	Tonne
Total Lightship Steel Weight Benchmark	**344.81**	**Tonne**

Schedule Table 66 Hull Girder Stress Verification Matrix (S.R.O. 308/Law-2001)

Stress Component	Governing Formula	Calculated	Statutory Limit	Compliance
Deck Bending Stress (σ_deck)	M_design / Z_deck	**51.76 N/mm²**	≤ 130.0 N/mm²	**COMPLIANT**
Bottom Bending Stress (σ_bottom)	M_design / Z_bottom	**30.16 N/mm²**	≤ 130.0 N/mm²	**COMPLIANT**
Hull Girder Shear Stress (τ)	Q_design / A_side_webs	**8.93 N/mm²**	≤ 70.0 N/mm²	**COMPLIANT**
Equivalent Combined Stress (σ_eq)	√(σ² + 3τ²)	**54.02 N/mm²**	≤ 150.0 N/mm²	**COMPLIANT**

Schedule Table 67 Local Structural Members & Buckling Audit (S.R.O. 308/Law-2001)

Local Structural Member	Governing Load Condition	Calculated	Allowable	Status
Bottom Transverse Floor Plate	Bottom hydrostatic head & cargo load (Rule 22)	**148.00 N/mm²**	150.0 N/mm²	**COMPLIANT**
Side Transverse Web Frame	Side hydrostatic pressure & dynamic wave action (Rule 26)	**66.34 N/mm²**	150.0 N/mm²	**COMPLIANT**
Deck Transverse Web Beam	Deck cargo & green water load (Rule 29)	**12.50 N/mm²**	150.0 N/mm²	**COMPLIANT**
Watertight Bulkhead Vertical Stiffener	Flooding head to bulkheads (Rule 37)	**120.54 N/mm²**	150.0 N/mm²	**COMPLIANT**
Underdeck Tubular Pillars	Axial deck cargo load & hold superstructure support (Rule 31)	**68.40 N/mm²**	120.0 N/mm²	**COMPLIANT**
Cargo Hatchway End Beam & Coaming	Hatch cover weight and deck wash load (Rule 50)	**22.00 N/mm²**	150.0 N/mm²	**COMPLIANT**
Rudder Stock & Pintle Assembly	Maximum steering torque at full ahead speed (Rule 40)	**82.50 N/mm²**	160.0 N/mm²	**COMPLIANT**

FINAL STATUTORY CERTIFICATION & APPROVAL

This is to certify that the structural scantlings, longitudinal bending moments, midship section modulus, and stress evaluations contained in this booklet have been audited and verified in accordance with the Bangladesh Inland Steel Ships Construction Rules 2001 (Rules 209 & 223), Schedule Tables 66 & 67, and ISO 1976 Inland Hull Strength Standard. The vessel meets all statutory longitudinal strength and section modulus requirements.

CHECKED & VERIFIED:	DESIGNED & SUBMITTED BY:
Engineer & Ship Surveyor Department of Shipping, Dhaka Date: _____	Naval Architect & Marine Structural Consultant Professional Registration No: _____ Date: _____

